4079. Prime Subset Selection I

Solved

Medium Hint

You are given two integers `n` and `s`.

Consider all **prime** numbers less than or equal to `n`. Select some of these primes such that:

- The **sum** of the selected primes is less than or equal to `s`.
- The number of selected primes is **maximized**.
- Each prime can be selected **at most once**.

Return an integer array containing the selected primes in **increasing** order. If multiple optimal selections exist, return **any** of them.

If no prime can be selected, return an **empty** array.

Example 1:

> **Input:** n = 7, s = 20
>
> **Output:** [2,3,5,7]
>
> **Explanation:**
>
> - The prime numbers less than or equal to `n = 7` are 2, 3, 5, and 7.
> - The sum of all these primes is `2 + 3 + 5 + 7 = 17`, which is less than or equal to `s = 20`.
> - Thus, the answer is `[2, 3, 5, 7]`.

Example 2:

> **Input:** n = 15, s = 15
>
> **Output:** [3,5,7]
>
> **Explanation:**
>
> - The prime numbers less than or equal to `n = 15` are 2, 3, 5, 7, 11, and 13.
> - One possible selection is `[3, 5, 7]`, whose sum is `3 + 5 + 7 = 15`.
> - The four smallest available primes have sum `2 + 3 + 5 + 7 = 17`, which exceeds `s = 15`. Therefore, selecting four or more primes is impossible, and the maximum number of selected primes is three.
> - Thus, `[3, 5, 7]` is one possible answer. Other valid answers include `[2, 3, 5]`, `[2, 3, 7]`, and `[2, 5, 7]`.

Example 3:

> **Input:** n = 5, s = 1
>
> **Output:** []
>
> **Explanation:**
>
> - The smallest prime number is 2, which is greater than `s = 1`.
> - Thus, no prime can be selected, so the answer is an empty array.

Constraints:

- $1 <= n <= 10^6$
- $1 <= s <= 10^9$

Seen this question in a real interview before? 1/6

Yes No

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Topics

Hint 1

Hint 2

Hint 3

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